Title

Cisco ENSA Enterprise Network Configuration Lab

Subtitle

Multi-Area OSPF, NAT, ACL, and Device Management Implementation

This repository contains the network topology and configuration scripts for a Cisco Enterprise Networking, Security, and Automation (ENSA) practice assessment. The objective was to configure a multi-site network (Ocisc LLC) implementing OSPFv2, NAT, ACLs, and device management.

Score Achieved: 88%

Project Summary: Cisco ENSA Enterprise Network Configuration

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Role: Network Security Engineer | Network Technician
Score: 88%

1. Executive Summary

This document serves as a technical report for a Cisco Enterprise Networking, Security, and Automation (ENSA) practice assessment. The objective was to design, configure, and secure a simulated multi-site enterprise network for a fictional company, Ocisc LLC. The project required the integration of dynamic routing, network address translation, traffic filtering, and device management to ensure seamless communication between the Eastern and Central branches.

2. Scope of Implementation

The configuration focuses on the following core networking principles:

- Multi-Area OSPFv2: Implemented to allow routers to dynamically share routes across the enterprise. Customizations included setting router IDs, adjusting timers, and configuring passive interfaces to secure edge ports from unnecessary routing updates.
- Network Address Translation (NAT): Configured both Static NAT (to allow external access to an internal server) and Dynamic NAT with PAT (Port Address Translation) to allow multiple internal hosts to share a limited pool of public IP addresses to access the internet.
- Access Control Lists (ACLs): Standard and Extended ACLs were deployed to enforce security policies. This included restricting VTY line access (Telnet/SSH) and blocking unauthorized HTTP traffic from specific LAN segments to critical servers.

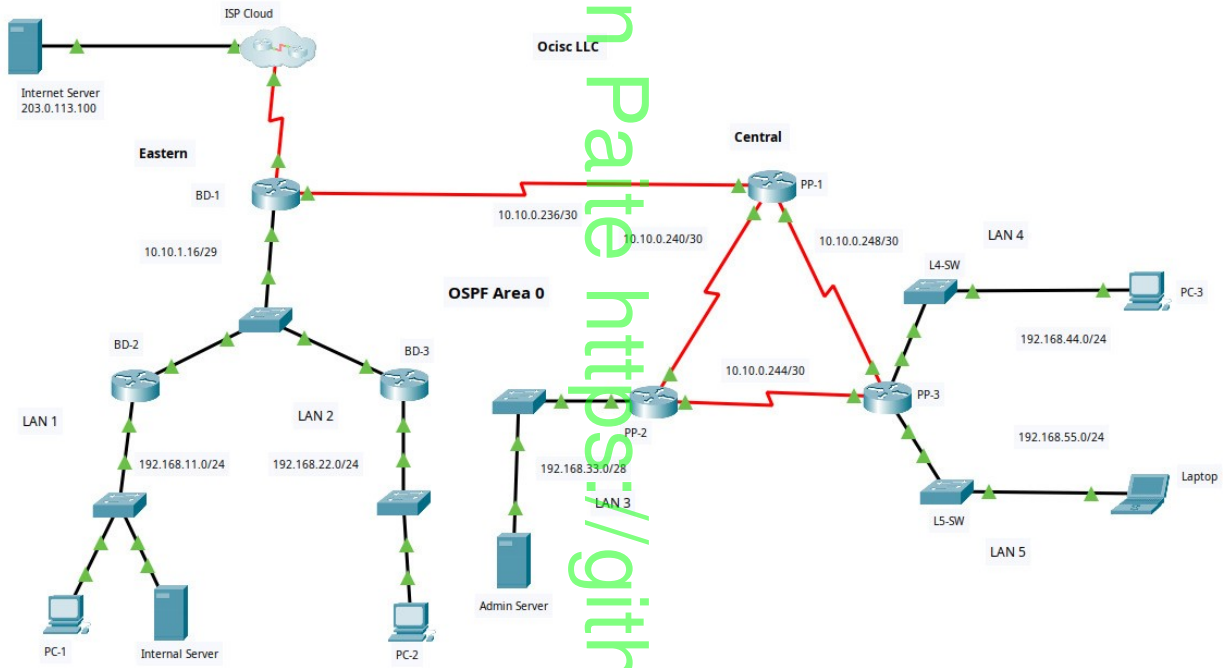
Giningakpio Stephen Paite https://github.com/paite404

- Network Management: Configured NTP (Network Time Protocol) to ensure accurate time synchronization across devices and performed a disaster recovery backup of the Cisco IOS image to a TFTP server.

3. Verification and Results

All configurations were tested and verified in Cisco Packet Tracer. Routing tables were checked for OSPF adjacencies, NAT translations were confirmed via `show ip nat translations`, and ACLs were verified to ensure they were filtering traffic as intended. The final assessment resulted in an 88% score, demonstrating a strong practical understanding of enterprise routing and security.

Network Topology



Technologies & Skills Implemented

- Routing: OSPFv2 (Multi-area, Router IDs, Interface-level activation, Auto-cost reference bandwidth, Hello/Dead timers).
- Security: Standard and Extended ACLs, VTY line restrictions.
- Translation: Static NAT and Dynamic NAT with PAT (Port Address Translation).
- Management: NTP configuration, TFTP IOS backup.

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Device Configurations

Below are the final running configurations for each device in the topology.

1. BD-1 (Eastern Router)

```
BD-1>
BD-1>
BD-1>enable
BD-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BD-1(config)#hostname BD-1
BD-1(config)#router ospf 10
BD-1(config-router)#router-id 9.9.9.9
BD-1(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BD-1(config-router)#default-information originate
BD-1(config-router)#passive-interface g0/0/0
BD-1(config-router)#exit
BD-1(config)#ip route 0.0.0.0 0.0.0.0 s0/1/1
%Default route without gateway, if not a point-to-point interface, may impact performance
BD-1(config)#interface g0/0/0
BD-1(config-if)#ip ospf 10 area 0
BD-1(config-if)#ip ospf priority 255
BD-1(config-if)#exit
BD-1(config)#interface s0/1/0
BD-1(config-if)#ip ospf 10 area 0
BD-1(config-if)#ip ospf hello-interval 20
BD-1(config-if)#ip ospf dead-interval 80
BD-1(config-if)#exit
BD-1(config)#interface s0/1/1
BD-1(config-if)#ip ospf 10 area 0
BD-1(config-if)#ip nat outside
BD-1(config-if)#exit
BD-1(config)#interface g0/0/0
BD-1(config-if)#ip nat inside
BD-1(config-if)#exit
BD-1(config)#ip nat inside source static 192.168.11.100 192.0.2.115
BD-1(config)#access-list 1 permit 192.168.0.0 0.0.255.255
BD-1(config)#ip nat pool POOL-1 192.0.2.116 192.0.2.118 netmask 255.255.255.248
BD-1(config)#ip nat inside source list 1 pool POOL-1 overload
BD-1(config)#ip access-list standard VTY-BLOCK
BD-1(config-std-nacl)#permit 192.168.0.0 0.0.255.255
BD-1(config-std-nacl)#deny any
BD-1(config-std-nacl)#exit
BD-1(config)#line vty 0 4
BD-1(config-line)#access-class VTY-BLOCK in
BD-1(config-line)#exit
BD-1(config)#end
BD-1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

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2. BD-2 (Eastern Router)

```
BD-2>
BD-2>enable
BD-2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BD-2(config)#hostname BD-2
BD-2(config)#router ospf 10
BD-2(config-router)#router-id 8.8.8.8
BD-2(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BD-2(config-router)#passive-interface g0/0/0
BD-2(config-router)#exit
BD-2(config)#interface g0/0/0
BD-2(config-if)#ip ospf 10 area 0
BD-2(config-if)#exit
BD-2(config)#interface g0/0/1
BD-2(config-if)#ip ospf 10 area 0
BD-2(config-if)#exit
BD-2(config)#end
BD-2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
BD-2#
BD-2#
```

3. BD-3 (Eastern Router)

```
BD-3>
BD-3>enable
BD-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BD-3(config)#hostname BD-3
BD-3(config)#router ospf 10
BD-3(config-router)#router-id 7.7.7.7
BD-3(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BD-3(config-router)#passive-interface g0/0/0
BD-3(config-router)#exit
BD-3(config)#access-list 10 deny 192.168.11.0 0.0.0.255
BD-3(config)#access-list 10 permit any
BD-3(config)#interface g0/0/0
BD-3(config-if)#ip ospf 10 area 0
BD-3(config-if)#exit
BD-3(config)#interface g0/0/1
BD-3(config-if)#ip ospf 10 area 0
BD-3(config-if)#ip access-group 10 in
BD-3(config-if)#exit
BD-3(config)#end
BD-3#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
BD-3#
BD-3#
08:05:28: %OSPF-5-ADJCHG: Process 10, Nbr 8.8.8.8 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

BD-3#
```

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4. PP-1 (Central Router)

```
BD-3>
BD-3>enable
BD-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BD-3(config)#hostname BD-3
BD-3(config)#router ospf 10
BD-3(config-router)#router-id 7.7.7.7
BD-3(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BD-3(config-router)#passive-interface g0/0/0
BD-3(config-router)#exit
BD-3(config)#access-list 10 deny 192.168.11.0 0.0.0.255
BD-3(config)#access-list 10 permit any
BD-3(config)#interface g0/0/0
BD-3(config-if)#ip ospf 10 area 0
BD-3(config-if)#exit
BD-3(config)#interface g0/0/1
BD-3(config-if)#ip ospf 10 area 0
BD-3(config-if)#ip access-group 10 in
BD-3(config-if)#exit
BD-3(config)#end
BD-3#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
BD-3#
BD-3#
08:05:28: %OSPF-5-ADJCHG: Process 10, Nbr 8.8.8.8 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

BD-3#
```

5. PP-2 (Central Router)

```
PP-2>
PP-2>enable
PP-2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
PP-2(config)#hostname PP-2
PP-2(config)#router ospf 10
PP-2(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
PP-2(config-router)#network 10.10.0.244 0.0.0.3 area 0
PP-2(config-router)#network 10.10.0.240 0.0.0.3 area 0
PP-2(config-router)#network 192.168.33.0 0.0.0.15 area 0
PP-2(config-router)#passive-interface g0/0/0
PP-2(config-router)#exit
PP-2(config)#ntp server 192.168.33.14
PP-2(config)#end
PP-2#write memory
%SYS-5-CONFIG_I: Configured from console by console

08:07:08: %OSPF-5-ADJCHG: Process 10, Nbr 10.10.0.249 on Serial0/1/1 from LOADING to FULL, Loading Done

Building configuration...
[OK]
PP-2#
```

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6. PP-3 (Central Router)

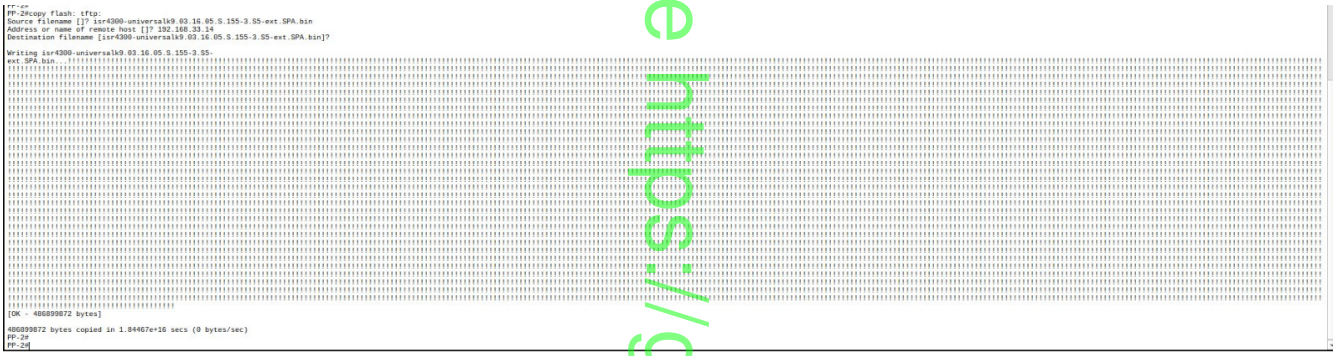
```
PP-3>
PP-3>
PP-3>enable
PP-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
PP-3(config)#hostname PP-3
PP-3(config)#router ospf 10
PP-3(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
PP-3(config-router)#network 10.10.0.248 0.0.0.3 area 0
PP-3(config-router)#network 10.10.0.244 0.0.0.3 area 0
PP-3(config-router)#network 192.168.44.0 0.0.0.255 area 0
PP-3(config-router)#network 192.168.55.0 0.0.0.255 area 0
PP-3(config-router)#passive-interface g0/0/0
PP-3(config-router)#passive-interface g0/0/1
PP-3(config-router)#exit
PP-3(config)#access-list 101 deny tcp 192.168.44.0 0.0.0.255 host 192.168.33.14 eq 80
PP-3(config)#access-list 101 permit ip any any
PP-3(config)#interface g0/0/0
PP-3(config-if)#ip access-group 101 in
PP-3(config-if)#exit
PP-3(config)#end
PP-3#write memory
%SYS-5-CONFIG_I: Configured from console by console

08:08:18: %OSPF-5-ADJCHG: Process 10, Nbr 10.10.0.249 on Serial0/1/0 from LOADING to FULL, Loading Done

08:08:19: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.33.1 on Serial0/1/1 from LOADING to FULL, Loading Done

Building configuration...
[OK]
PP-3#
PP-3#
```

FINAL STEP: IOS BACKUP (On PP-2)


PP-2#copy flash: tftp:
Source filename []? isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
Address or name of remote host []? 192.168.33.14
Destination filename [isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin]?
Writing isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin...
!!!